Example 1. Fit a straight line to the following data :

Year x	1961	1971	1981	1991	2001
Production y in tons	8	10	12	10	16

and find the expected production in 2006.

Soln.:- The normal equations from least square method for a straight line $y = ax + b$ is

$$a\sum_{i=1}^{n} x_i + nb = \sum_{i=1}^{n} y_i$$

$$a\sum_{i=1}^{n} x_i^2 + b\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} x_i y_i$$

Here $n = 5$. From the given data, we have, here we can consider $x = 1, 2, 3, 4, 5$

x	y	xy	x^2
1	8	8	1
2	10	20	4
3	12	36	9
4	10	40	16
5	16	80	25
$\sum_{i=1}^{7} x_i = 15$	$\sum_{i=1}^{7} y_i = 56$	$\sum_{i=1}^{7} x_i y_i = 184$	$\sum_{i=1}^{7} x_i^2 = 55$

Therefore the normal equations are given by:

$$15a + 5b = 56$$
$$55a + 15b = 184$$

On solving above equations, we get

$$a = 1.6 \quad \text{and} \quad b = 6.4$$

Hence the best straight line is $y = 1.6x + 6.4$.

The expected production in 2006 is obtained by taking $x = 5.5$, we get $y(2006) = 15.2$ tons.

Example 2. The results of measurements of electric resistance R of a copper bar at various temperatures $t^{\circ}C$ are listed below :

t	19	25	30	36	40	45	50
R	76	77	79	80	82	83	85

Find a relation $R = a + bt$ where a and b are constants to be determined.

Soln.:- Consider the normal equations of least square fit of a straight line i.e.

$$an + b\sum_{i=1}^{n} t_i = \sum_{i=1}^{n} R_i$$

$$a\sum_{i=1}^{n} t_i + b\sum_{i=1}^{n} t_i^2 = \sum_{i=1}^{n} t_i R_i$$

Here $n = 7$. From the given data, we have,

t	R	t^2	tR
19	76	361	1444
25	77	625	1925
30	79	900	2370
36	80	1296	2880
40	82	1600	3280
45	83	2025	3735
50	85	2500	4250
$\sum_{i=1}^{7} t_i = 245$	$\sum_{i=1}^{7} R_i = 562$	$\sum_{i=1}^{7} t_i^2 = 9307$	$\sum_{i=1}^{7} t_i R_i = 19884$

Therefore the normal equations are given by:

$$7a + 245b = 562$$
$$245a + 9307b = 19884$$

On solving above equations, we get

$$a = 70.0535 \quad \text{and} \quad b = 0.2923$$

Hence the best relation between t and R is $R = 70.0535t + 0.2923$.

Example 3. If P is the pull required to lift a load W by means of a pulley block, find a linear law of the form $P = mW + C$, connecting P and W,using the following data

P(in Kg)	12	15	21	25
W(in Kg)	50	70	100	120

Use least square method to compute P when W =150kg.

Soln.:- The normal equations from least square method for a straight line $P = mW + C$ is

$$m\sum_{i=1}^{n} W_i + nC = \sum_{i=1}^{n} P_i$$

$$m\sum_{i=1}^{n} W_i^2 + C\sum_{i=1}^{n} W_i = \sum_{i=1}^{n} W_i P_i$$

Here $n = 4$. From the given data, we have,

P	W	W^2	WP
12	50	144	600
15	70	225	1050
21	100	441	2100
25	120	625	3000
$\sum_{i=1}^{7} P_i = 73$	$\sum_{i=1}^{7} y_i = 340$	$\sum_{i=1}^{7} x_i y_i = 31800$	$\sum_{i=1}^{7} x_i^2 = 6750$

Therefore the normal equations are given by:

$$340m + 4C = 73$$

$$31800m + 340C = 6750$$

On solving above equations, we get

$$m = 0.1879 \quad \text{and} \quad C = 2.2759$$

Hence the best relation between P and W is $P = 0.1879W + 2.2759$.

The value of P when $W = 150$ is approximately 30.46 Kg.

Fitting a Parabola (Quadratic Function)

Let $y = f(x) = ax^2 + bx + c$ be the parabola to be fitted to the given set of n data points $(x_1, y_1), (x_2, y_2), (x_3, y_3), \cdots, (x_n, y_n)$.

The error between the data points and the parabola is

$$e_i = y_i - f(x_i) = y_i - (ax_i^2 + bx_i + c), \quad i = 1,2,3, \cdots, n$$

The sum of the squares of the errors is

$$E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - (ax_i^2 + bx_i + c)\right)^2.$$

Now by the method of least square, for the curve of best fit, E is minimum
That is,

$$\frac{\partial E}{\partial a} = \frac{\partial E}{\partial b} = \frac{\partial E}{\partial c} = 0.$$

From the above equation, we get normal equations as:

$$a\sum_{i=1}^{n} x_i^2 + b\sum_{i=1}^{n} x_i + cn = \sum_{i=1}^{n} y_i$$

$$a\sum_{i=1}^{n} x_i^3 + b\sum_{i=1}^{n} x_i^2 + c\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} x_i y_i$$

$$a\sum_{i=1}^{n} x_i^4 + b\sum_{i=1}^{n} x_i^3 + c\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i^2 y_i$$

After solving above normal equations we get values of a, b and c.

Example 1. Fit a parabola $y = ax^2 + bx + c$ to the given data

x	10	12	15	20	23
y	14	17	23	21	25

Soln.: Let the parabola of best fit be given by

$$y = ax^2 + bx + c$$

Normal equations are given by:

$$a\sum_{i=1}^{n} x_i^2 + b\sum_{i=1}^{n} x_i + cn = \sum_{i=1}^{n} y_i$$

$$a\sum_{i=1}^{n} x_i^3 + b\sum_{i=1}^{n} x_i^2 + c\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} x_i y_i$$

$$a\sum_{i=1}^{n} x_i^4 + b\sum_{i=1}^{n} x_i^3 + c\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i^2 y_i$$

x	y	xy	x^2	x^2y	x^3	x^4
10	14	140	100	1400	1000	10000
12	17	204	144	2448	1728	20736
15	23	345	225	5175	3375	50625
20	21	420	400	8400	8000	160000
23	25	575	529	13225	12167	279841
$\sum_{i=1}^{n} x_i = 80$	$\sum_{i=1}^{n} y_i = 100$	$\sum_{i=1}^{n} x_i y_i = 1684$	$\sum_{i=1}^{n} x_i^2 = 1398$	$\sum_{i=1}^{n} x_i^2 y_i = 30648$	$\sum_{i=1}^{n} x_i^3 = 26270$	$\sum_{i=1}^{n} x_i^4 = 521202$

Substituting these values into normal equations, we get

$$1398a + 80b + 5c = 100$$
$$26270a + 1398b + 80c = 1684$$
$$521202a + 26270b + 1398c = 30648$$

After solving above linear systems, we get $a = -0.0695, b = 3.0099, c = -8.7279.$

∴ The parabola of best fit is $y = -0.0695x^2 + 3.0099x - 8.7279.$

Example 2. Fit a parabola of the form $y = a + bx + cx^2$ which fit most closely with the observations:

x	-3	-2	-1	0	1	2	3
y	4.63	2.11	0.67	0.09	0.63	2.15	4.58

Evaluate y at $x = 0.5$ and $x = 1.5$

Soln.:- Let the parabola of best fit be given by

$$y = a + bx + cx^2$$

Normal equations are given by:

$$an + b\sum_{i=1}^{n} x_i + c\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} y_i$$
$$a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} x_i^2 + c\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i y_i$$
$$a\sum_{i=1}^{n} x_i^2 + b\sum_{i=1}^{n} x_i^3 + c\sum_{i=1}^{n} x_i^4 = \sum_{i=1}^{n} x_i^2 y_i$$

x	y	xy	x^2	x^2y	x^3	x^4
-3	4.63	-13.89	9	41.67	-27	81
-2	2.11	-4.22	4	8.44	-8	16
-1	0.67	-0.67	1	0.67	-1	1
0	0.09	0	0	0	0	0
1	0.63	0.63	1	0.63	1	1
2	2.15	4.3	4	8.6	8	16
3	4.58	13.74	9	41.22	27	81
$\sum_{i=1}^{n} x_i = 0$	$\sum_{i=1}^{n} y_i = 14.86$	$\sum_{i=1}^{n} x_i y_i = -0.11$	$\sum_{i=1}^{n} x_i^2 = 28$	$\sum_{i=1}^{n} x_i^2 y_i = 101.23$	$\sum_{i=1}^{n} x_i^3 = 0$	$\sum_{i=1}^{n} x_i^4 = 196$

Substituting these values into normal equations, we get

$$7a + 0b + 28c = 14.86 \Rightarrow a + 4c = 2.1229$$

$$0a + 28b + 0c = -0.11 \Rightarrow b = -0.0039$$

$$28a + 0b + 196c = 101.23 \Rightarrow a + 7c = 3.6154$$

After solving above linear systems, we get $a = 0.1329, b = -0.0039, c = 0.4975$

$\therefore$ The parabola of best fit is $y = 0.1329 - 0.0039x + 0.4975x^2$.

$$\therefore y(0.5) = 0.2553 \quad \text{and} \quad y(1.5) = 1.2463.$$